



Lesson 1.3 — Geometric Sequences

Big idea: A geometric sequence multiplies by the same number (the common ratio r) every step. That's the doorway to exponential thinking.

Quick review

Common ratio r = any term divided by the one before it. If r is constant (and nonzero), the sequence is geometric.

Recursive: a_1 given, $a_n = r \cdot a_{n-1}$.

Explicit: $a_n = a_1 \cdot r^{(n-1)}$.

Connection to exponentials: if you replace the integer index n with a continuous variable, you get an exponential function. That's why geometric sequences show up in growth/decay problems.

Worked example

Worked example. A sequence starts 3, 6, 12, 24, ... Find the 10th term.

$$r = 6/3 = 2, a_1 = 3. a_{10} = 3 \cdot 2^9 = 3 \cdot 512 = 1,536.$$

Warm-ups

1. For the sequence 4, 12, 36, 108, ..., state r and write the explicit rule.
2. For the sequence 80, 40, 20, 10, ..., state r and find a_7 .
3. Is 1, 4, 9, 16, ... geometric? Explain in one sentence.

Practice

1. A geometric sequence has $a_1 = 5$ and $r = 3$. Find a_8 .
2. A geometric sequence has $a_2 = 18$ and $a_5 = 486$. Find r and a_1 .
3. A bacteria colony doubles every hour. If it starts at 200 bacteria, how many are there after 8 hours?

Challenge

1. A car is worth \$28,000 new and loses 15% of its value each year. Write an explicit rule for its value at year n (with year 1 = new). What is it worth after 6 years?
2. Compare these two sequences that start at 1 at $n=1$:
A: arithmetic with $d = 3$ (1, 4, 7, ...)
B: geometric with $r = 1.1$ (1, 1.1, 1.21, ...)
Which is larger at $n = 10$? At $n = 50$? Explain why the answer flips.



Answer key — Lesson 1.3

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $r = 3$; $a_n = 4 \cdot 3^{(n-1)}$.
2. $r = 0.5$; $a_7 = 80 \cdot (0.5)^6 = 1.25$.
3. No — the ratios are $4/1$, $9/4$, $16/9$, not constant. (It's n^2 , a quadratic sequence.)

Practice

1. $a_8 = 5 \cdot 3^7 = 10,935$.
2. $r^3 = 486/18 = 27 \Rightarrow r = 3$. $a_1 = 18/3 = 6$.
3. $200 \cdot 2^8 = 51,200$ bacteria.

Challenge

1. $a_n = 28,000 \cdot (0.85)^{(n-1)}$. Year 6: $28,000 \cdot 0.85^5 \approx \$12,424$.
2. At $n=10$: $A = 28$, $B \approx 2.36 \rightarrow A$ is larger. At $n=50$: $A = 148$, $B \approx 106.7 \rightarrow$ still $A \dots$ but at $n \approx 95$ they cross. Exponentials (even with small r) eventually overtake any arithmetic sequence because they multiply rather than add.